

Excitation-Aware On-Track System Identification at Full Scale: An Identifiability Analysis and a Simulation Architecture for Pacejka Estimation

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Abstract—On-track system identification promises tire models obtained from ordinary racing laps instead of dedicated steady-state manoeuvres. This paper studies the identifiability of one such pipeline—the residual-network plus virtual-steady-state-data method of Dikici *et al.* [1]—at full vehicle scale, and reports a corrected and extended identification pipeline together with the simulation architecture that makes it verifiable. The plant is a Formula Student vehicle simulated in CARLA and driven through a purpose-built ROS 2 bridge that exposes the simulator’s own per-wheel slip angle, normal load and lateral force, so that an on-track estimator can be scored against the forces the plant actually generated—a comparison that a physical vehicle cannot provide without a force-measuring hub. Transplanted unchanged, the pipeline converges on a tire whose peak grip is 0.40 *g*, with five of eight Pacejka coefficients resting on their box constraints, for a vehicle that measures 1.00–1.05 *g*. We show that this is neither a solver, a bounds nor a tuning failure but a loss of excitation created by the pipeline’s own virtual-data step: the friction the synthetic rollout can demand is $\mu_{\text{reach}} = v^2 \delta_{\text{sweep}} / (gL)$, a property of the rollout configuration and not of the tire, capped at 0.43 by a rollout speed inherited from a 1:10 platform. The paper contributes (i) this identifiability bound and an excitation-aware rollout that removes the degeneracy; (ii) an extrapolation bound on the synthetic steering sweep together with a frozen-prior Tikhonov term, which jointly stop the peak factor from drifting across co-identification iterations; (iii) three measurement-side corrections (achieved rather than commanded road-wheel angle, measured mass and geometry, and a peak-rather-than-median friction warm-start applied as a floor); (iv) a ROS 2 simulation architecture that closes the identification loop end to end, from data acquisition through fitting to ground-truth force validation; and (v) admissibility gates on derived physical quantities, placed at the model-to-controller boundary. After correction, three consecutive live identifications return no coefficient on a bound, peak friction coefficients of 1.05–1.18, axle cornering stiffnesses of 21 kN/rad front and 20 kN/rad rear against a simulator ground truth of 17–19 kN/rad, and one-step prediction R^2 of 0.97 for lateral velocity and 0.99 for yaw rate. The corrected identifier also makes a negative result visible: the optimised raceline used in this study demands 1.22 *g* and is therefore infeasible for this vehicle, and the admissibility gate rejects deployment—correctly.

Index Terms—System identification, tire modeling, Pacejka Magic Formula, identifiability, persistent excitation, vehicle dynamics, autonomous racing, simulation architecture, CARLA, ROS 2.

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All experiments reported in this paper are performed in simulation, using the CARLA simulator and the CARLA_ASU_BRIDGE ROS 2 interface described in Section IV.

I. INTRODUCTION

MODEL-BASED controllers for autonomous racing derive their performance from the fidelity of the vehicle model they carry [2], [3], [4]. In the lateral channel that fidelity is dominated by the tire, which is simultaneously the physical bound on manoeuvrability and the least accessible component to measure: its behaviour depends on surface, temperature, wear and load in ways that change between sessions [5], [6].

Classical characterisation resolves this by removing the vehicle from the track—steady-state circular or ramp-steer manoeuvres in a large open space yield clean data at a logistical cost a race weekend rarely affords [1]. On-track identification removes the logistics but reintroduces transients, noise and, critically, the fact that a racing line is a poor excitation signal for a nonlinear model; nonlinear least squares applied directly to lap data is known to be brittle in exactly this regime [1].

Dikici *et al.* [1] propose the hybrid remedy that is the starting point of this paper. A small residual network is trained on 30 s of ordinary lap data to predict the one-step error of a nominal single-track model; the corrected model is then rolled out through a *synthetic*, quasi-steady steering sweep; and Pacejka coefficients are fitted to that synthetic sweep. Because the sweep is generated rather than driven it can be made steady and noise-free, and because the residual network only has to interpolate within its training distribution its generalisation weakness is contained. The procedure iterates, each fit becoming the next nominal model. On a 1:10 platform it reports a $3.3\times$ lower one-step RMSE than nonlinear least squares under noise, from 30 s of driving [1].

A. Problem statement

This paper asks what the pipeline identifies when it is moved from a 3.7 kg, 0.32 m-wheelbase platform to a full-scale Formula Student vehicle: 269.6 kg, 1.53 m wheelbase, 16° of road-wheel lock, driven at 11–27 m/s. The question is one of identifiability rather than of parameter transfer. Two scale-dependent mechanisms, both latent at 1:10, break the estimate:

- 1) **Excitation collapses inside the virtual-data step.** The Pacejka fit never sees the recorded lap; it sees the synthetic rollout, whose reachable friction $\mu_{\text{reach}} \approx v^2 \delta_{\text{max}} / (gL)$ is a property of the rollout configuration, not of the tire. A rollout speed inherited from the scaled platform caps it at 0.43 here. Below the tire’s peak

the Magic Formula degenerates to $F_y \approx F_z(BCD)\alpha$: only the product BCD is identifiable and the bounded optimiser resolves the remaining freedom on a box edge [7], [5].

- 2) **Lateral demand scales as v^2 , so a bad grip estimate is silent until it is catastrophic.** An identified peak friction coefficient of 0.40 produces 0.03 m of cross-track error at 11 m/s and 41.89 m at 27 m/s. The scaled platform never reaches the speed at which the error becomes observable.

A third difference is an opportunity. Because the study is conducted in the CARLA simulator [8] through a purpose-built ROS 2 interface, the simulator’s own per-wheel telemetry—slip angle, normal load, lateral force, per wheel—is available as ground truth, so an on-track identifier can be scored against the forces the plant actually generated.

B. Contributions

The contributions are an identifiability result, the pipeline changes it implies, and the simulation architecture that makes both verifiable.

- 1) **An identifiability analysis of the virtual-data step** (Section V-B), giving the closed-form reachable-friction bound above and an *excitation-aware rollout* that selects the rollout speed from the measured lap and warns whenever the configuration cannot reach the prior’s peak. This corrects the baseline method [1] at any scale; it is latent at 1:10 because a 2–4 m/s rollout is representative there and is not here.
- 2) **Two mechanisms that stabilise the iterative co-identification** (Section V-C): the synthetic sweep is bounded at $1.5\times$ the 99th percentile of the observed steering, so the fit cannot read the residual network’s extrapolation as grip; and the Tikhonov target is frozen at the configured prior while the optimiser’s start point tracks the iterations. Together they remove a $1.01\rightarrow 1.87$ drift of the peak factor over six iterations.
- 3) **Measurement-side corrections for full-scale deployment** (Section V-E): slip angles built from the *achieved* road-wheel angle rather than the command; mass and wheelbase from measured static wheel loads rather than from a configuration file; and a friction warm-start computed as a high quantile over the whole buffer and applied as a *floor* rather than as a median over a low-slip mask.
- 4) **A simulation architecture for on-track identification research** (Sections IV-B–IV-D). CARLA_ASU_BRIDGE exposes the plant’s own per-wheel forces, slip angles and loads as ROS 2 topics alongside a conventional autonomy interface, and closes the identification loop end to end—acquisition, preprocessing, fitting, ground-truth force validation—on a single simulation clock.
- 5) **Admissibility gates on derived physical quantities** (Section VII) at the model-to-controller boundary—axle cornering stiffness, understeer sign and critical speed, and grip ceiling against the trajectory’s lateral demand—rather than per-coefficient box checks, which cannot

detect a degenerate fit or a pairwise-oversteering axle set.

- 6) **A negative result reported rather than suppressed** (Section VI-G). The corrected model reports a grip ceiling of 1.02–1.16 g against a measured steady-state ceiling of $\approx 1.0 g$, while the optimised raceline used here demands 1.22 g ; the gate refuses the identification for closed-loop use, and we quantify the raceline regeneration this implies instead of loosening the gate.

The paper follows that order: Section II places the work against the state of the art; Section III states the identification problem and its structural identifiability properties; Section IV describes the simulation architecture and data pipeline; Section V the identification methodology; Section VI the results; Section VII how the identified model is handed to the controller; and Sections VIII–IX the limitations and outlook. All results are obtained in simulation, and Section VIII states which conclusions are expected to transfer to hardware.

II. RELATED WORK AND POSITION

A. Tire models and Magic Formula identification

The Magic Formula of Bakker, Nyborg and Pacejka [9] and its consolidation [10], [5] remains the standard empirical description of steady-state tire force generation and is the model identified here; its lateral form and the two structural properties this paper relies on are stated in Section III. Classical characterisation obtains the coefficients from steady-state manoeuvres in a controlled space [1], [6]: clean data at high logistical cost. Direct fitting to measured contact-patch forces is the most informative variant and requires a force-measuring hub or a simulator’s own wheel telemetry; in this work it is used only as a *reference* against which the on-track identifier is scored, never inside the identification loop.

B. On-track and data-driven identification

On-track methods trade excitation quality for operational simplicity. Nonlinear least squares on lap data is the classical baseline and is brittle under noise [1]. Gaussian-process approaches learn the mismatch of a nominal model and have been used both for identification and inside the controller [11], [12], [3]; they are accurate but carry a continuing computational cost and, as [1] observes, are usually evaluated starting from an already accurate physical model. Neural residuals replace the Gaussian process with a cheaper regressor at the cost of generalisation guarantees.

Deep Dynamics [13] is the closest work on the physics-constraint axis: it estimates Pacejka coefficients inside a neural network and adds a *Physics Guard* layer clamping the internal estimates into nominal ranges. Our admissibility gates share the motivation but differ in placement and in what is constrained. We report as a substantive finding that per-coefficient box constraints are *insufficient*: the degenerate fit that motivated this work satisfied every box constraint while five of eight coefficients sat exactly on a bound, and a later fit satisfied every box constraint while being pairwise oversteering and open-loop unstable above 17.4 m/s. We therefore

gate on derived quantities at the model-to-controller interface and treat a coefficient resting on a bound as a diagnostic of non-identifiability rather than as a successful clamp.

The hybrid residual-plus-virtual-data structure of [1] is the pipeline we extend. Its key idea—query the learned model only where it interpolates, and extract physically parameterised coefficients from it—is retained unchanged. What we correct is the mechanism that decides *where* the learned model is queried.

C. Overlap with concurrent work

Wu *et al.* [14] extend the same baseline concurrently and independently, and the overlap must be stated plainly. They introduce (i) a vision-based friction prior from a lightweight CNN over road texture, used to warm-start the optimisation; (ii) an S4 structured state-space model in place of the MLP; and (iii) Nelder–Mead for the iterative Pacejka extraction, with virtual simulation in CarSim. The pipeline described here independently contains an S4D temporal residual [15], [16], a friction warm-start, and Nelder–Mead and differential-evolution solver options alongside trust-region reflective. We therefore do *not* claim the temporal-residual architecture, the friction warm-start concept, or the derivative-free solver as contributions.

Two differences are technical rather than chronological. Our friction prior is proprioceptive—derived from measured accelerations, in the spirit of model-free friction estimation from inertial data [17]—and therefore needs no camera or textured-surface assumption. More importantly, a prior computed from *utilised* friction is a lower bound on *available* friction and cannot substitute for excitation: on a gentle lap the median utilisation on our vehicle measures 0.044. We consequently apply the prior as a floor and locate the actual fix in the rollout excitation rather than in the initialisation. To our knowledge the reachable-friction bound of Section V-B and its consequence for the virtual-data step have not previously been stated.

D. Simulation architectures for identification and validation

CARLA [8] is the standard open simulator for autonomous-driving research and supplies the sensor and scenario infrastructure used here. Its native dynamics come from the underlying PhysX wheeled-vehicle model—a rigid-body model with per-wheel tire friction rather than an empirical Magic Formula fit to measured data—which is why this study treats the simulator as a well-instrumented *plant* to be identified and not as a source of ground-truth Pacejka coefficients. Recent work addresses that fidelity gap directly: R-CARLA [18] makes the dynamics model interchangeable while retaining CARLA’s sensor simulation, and Sagmeister *et al.* [19] quantify how far a vehicle model can be simplified before conclusions about a trajectory-following controller stop transferring—finding, relevantly here, that sensitivity is largest near the acceleration limits. CARLA has also hosted complete Formula Student Driverless stacks with ROS 2 interfaces [20]. The architecture of Section IV sits in this line and is distinguished by exposing the simulator’s own per-wheel slip, load and force

telemetry on a simulation-time-stamped interface, so that an identification method can be scored against the forces the plant actually produced rather than against a secondary model. Scaled platforms—F1TENTH [21] and the ForzaETH Race Stack [4]—provide the software context of the baseline; full-scale Formula Student Driverless systems [6] the vehicle class targeted here.

E. Downstream consumer and gap addressed

The identified model is consumed by a nonlinear MPC path-tracking controller built on *acados* [22] with partial-condensing HPIPM [23], following standard practice in racing MPC [2], [3], tracking a minimum-curvature raceline generated in the manner of Heilmeier *et al.* [24]. We claim no contribution to the controller itself; the controller appears in this paper only as the consumer that defines what an admissible identification is (Section VII).

Summarising the gap: the baseline [1] establishes that a residual network plus virtual steady-state data recovers a Pacejka model from a lap at 1:10 scale, and concurrent work [14] improves its residual architecture and its initialisation. Neither addresses the condition under which the virtual data itself carries information about the tire’s peak, which is the binding constraint at full scale, and neither treats the admissibility of the identified model for its downstream controller as an explicit, physically derived contract. This paper addresses both, on a platform where the answer can be checked against the plant’s own forces.

III. IDENTIFICATION PROBLEM

A. Model structure

The identification model is the dynamic single-track model of the baseline, with mass m , yaw inertia I_z and centre-of-gravity distances l_f , l_r , $L = l_f + l_r$. Longitudinal and lateral dynamics are decoupled and load transfer is neglected:

$$\dot{v}_y = \frac{1}{m}(F_{y,r} + F_{y,f} \cos \delta - mv_x \omega), \quad (1)$$

$$\dot{\omega} = \frac{1}{I_z}(F_{y,f} l_f \cos \delta - F_{y,r} l_r), \quad (2)$$

with slip angles

$$\alpha_f = \delta - \arctan\left(\frac{v_y + l_f \omega}{v_x}\right), \quad \alpha_r = -\arctan\left(\frac{v_y - l_r \omega}{v_x}\right). \quad (3)$$

Axle forces follow the lateral Magic Formula

$$F_y = F_z D \sin(C \arctan(B\alpha - E(B\alpha - \arctan B\alpha))), \quad (4)$$

with per-axle coefficients $\Phi_p = [B_f, C_f, D_f, E_f, B_r, C_r, D_r, E_r]$. The identification state and input are $\mathbf{x} = [v_y, \omega]$ and $\mathbf{u} = [v_x, \delta]$, and the nominal one-step map is $\hat{\mathbf{x}}_{k+1} = f(\mathbf{x}_k, \mathbf{u}_k, \Phi_p)$. The baseline discretises (1)–(2) by explicit Euler at T_s ; we retain Euler as the default and expose second- and fourth-order Runge–Kutta as options, because the full-scale vehicle’s lateral mode is stiffer relative to T_s than the scaled platform’s.

B. Objective and derived quantities

The identification objective is to recover Φ_p from a single 30 s segment of ordinary lap data, without dedicated manoeuvres, such that the resulting model is both predictive and *admissible* for the controller that consumes it. Admissibility is expressed on derived quantities rather than on coefficients. The linearised axle cornering stiffness is the slope of (4) at zero slip,

$$C_i = F_{z,i} B_i C_i^{\text{MF}} D_i, \quad i \in \{f, r\}, \quad (5)$$

(writing C^{MF} for the shape factor to avoid a clash), and the grip ceiling implied by an identified set is

$$a_y^{\text{max}} = \frac{D_f F_{z,f} + D_r F_{z,r}}{m}. \quad (6)$$

The linear model built from (5) is oversteering when $l_f C_f > l_r C_r$ and is then open-loop unstable above the critical speed $v_{\text{crit}} = L \sqrt{C_f C_r / (m(l_f C_f - l_r C_r))}$ [25].

C. Structural identifiability

Two properties of (4) are used throughout. First, its slope at zero slip is $F_z BCD$: the cornering stiffness depends on the coefficients only through that product, and E drops out. Second, and decisively, if the data does not approach the peak then (4) linearises to $F_y \approx F_z (BCD) \alpha$ and B, C, D become mutually unidentifiable. This is the standard persistent-excitation argument [7] and is not claimed as novel. What we claim, and demonstrate in Section V-B, is that the baseline pipeline's virtual-data step can silently violate it, and that a bounded optimiser then resolves the remaining freedom by sliding onto a box edge—so the violation is reported as a converged fit rather than as a failure.

D. Baseline pipeline

The baseline proceeds in four stages, repeated for N_{it} iterations [1]:

- 1) **Residual learning.** A small network f_{NN} is trained on the recorded lap to predict the one-step error $e_k = \mathbf{x}_{k+1} - \hat{\mathbf{x}}_{k+1}$ from $[v_x, v_y, \omega, \delta]_k$; the corrected model is $\mathbf{x}_{k+1} = \hat{\mathbf{x}}_{k+1} + e_k$.
- 2) **Virtual steady-state generation.** The corrected model is integrated open-loop at constant longitudinal speed while the steering angle is ramped from 0 to δ_{sweep} , producing a synthetic quasi-steady dataset.
- 3) **Pacejka fit.** Assuming $\dot{v}_y = \dot{\omega} = 0$ on the synthetic data, axle forces are recovered from the yaw balance,

$$F_{y,r} = \frac{m l_f v_x \omega}{L}, \quad F_{y,f} = \frac{m l_r v_x \omega}{L \cos \delta}, \quad (7)$$

slip angles from (3), and Φ_p is fitted to $(\alpha_i, F_{y,i})$ by bounded least squares.

- 4) **Re-nomination.** The fitted Φ_p becomes the next iteration's nominal model and the residual network is reinitialised.

The structural point that drives the rest of this paper is that stage 3 *never sees the recorded lap*. It sees the output of stage 2. The recorded lap enters only through the residual network, so the excitation available to the fit is set by the rollout configuration and not by how the vehicle was driven.

IV. SIMULATION ARCHITECTURE AND DATA PIPELINE

The identification pipeline is exercised end to end inside a single simulation architecture: a Formula Student vehicle model in CARLA acting as the plant, a C++ ROS 2 bridge that drives it and republishes its state, and an identification stack that consumes that state. The architecture is described here because two of its properties—per-wheel ground truth and simulation-time stamping—are what make the identifiability claims of Section V checkable rather than merely arguable.

A. Choice of simulator

CARLA [8] was selected against four requirements that the identification problem of Section III imposes on a simulator.

Per-wheel ground truth. The claim to be checked is that the recovered Magic Formula reproduces the lateral force the plant actually produced, so the simulator must expose per-wheel slip angle, normal load and tire force—not only chassis states. CARLA's PhysX wheeled-vehicle solver computes these quantities internally and its client API exports them, which is what makes the `feedback/tire_forces` channel of Section IV-C possible. Cone-oriented Formula Student simulators built on the AirSim plugin [26] publish a sensor-level interface (IMU, GNSS, LiDAR, camera) but no per-wheel force telemetry, and on hardware the same signal would require force-measuring hubs.

Deterministic, simulation-time-stamped execution. Persistent-excitation and residual-learning arguments are only checkable if every sample carries a consistent clock. CARLA's synchronous mode with a fixed physics step (0.033 s here) decouples data timing from wall-clock load; all identification data is stamped from the simulation clock (Appendix, Section A7).

Full-scale vehicle and racing-track support with a native ROS 2 path. The target class is a full-scale Formula Student car on a closed circuit. CARLA accepts custom vehicle assets—here `vehicle.vehicle.asurt_fsai` on the `silverstone` map—and offers a C++ client that the bridge links against directly, avoiding a Python round trip in the control loop. Complete Formula Student Driverless stacks have already been built on this combination [20]. Surveys of open self-driving simulators place CARLA (with the now-discontinued LGSVL) at the state of the art for end-to-end stack testing on exactly these grounds [27].

A plant whose tire behaviour is not a Magic Formula. CARLA's native dynamics are a known fidelity gap: R-CARLA [18] replaces the dynamics model to shrink a measured sim-to-real gap, and Sagmeister *et al.* [19] quantify where model simplification changes control conclusions. For this study that gap is a feature rather than a defect. Because forces come from a friction-based rigid-body solver and not from an analytic tire model, the simulator has no built-in Pacejka coefficients for the identifier to trivially recover; the reference curve must itself be fitted to the plant's telemetry (Section VI), which is the same position the method occupies on a real vehicle. The limits of this choice are revisited in Section VIII.



Fig. 1. The Formula AI vehicle model, `vehicle.vehicle.asurt_fsai`, on the start/finish straight of the silverstone map in CARLA. Its lateral forces are produced by the engine’s PhysX wheeled-vehicle solver rather than by an analytic tire model, which is why the reference curve of Table III is fitted to the simulator’s own per-wheel telemetry. Measured geometry and mass are given in Table I.

TABLE I
VEHICLE CONFIGURATION AND MEASURED PLANT PROPERTIES.
CONFIGURED VALUES ARE SET IN THE BRIDGE’S `CARLA_INTERFACE_CONFIG.YAML`; MEASURED VALUES ARE OBTAINED FROM THE SIMULATOR’S PER-WHEEL TELEMETRY.

Quantity	Value	Unit
Mass m	240.0	kg
Static wheel load (total)	2644.5	N
Front/rear static split	51.2 / 48.8	%
Wheelbase L	1.5334	m
l_f / l_r	0.7381 / 0.7953	m
Yaw inertia I_z^a	51.1	kg m ²
Wheel radius	0.25	m
Max road-wheel angle	16.0	°
PhysX tire_friction	1.5	—
Realised peak axle μ	1.00–1.05	—
Peak transient a_y	1.50	g
Steady-state a_y ceiling	≈ 1.0	g
Axle cornering stiffness	$\approx 12.1 F_z$	N/rad
front / rear	17198 / 16516	N/rad
Drag coefficient	0.426	—

^aAssumed; see Section A.

B. Plant: Formula AI vehicle in CARLA

The plant is the driverless Formula Student-class vehicle model `vehicle.vehicle.asurt_fsai`, instantiated in CARLA [8] (server 0.9.16, Unreal Engine 4.26) on the silverstone map: a four-wheeled, front-steered, all-wheel-drive rigid-body vehicle whose lateral forces are generated by the PhysX wheeled-vehicle solver under a configured friction coefficient, not by an analytic Magic Formula. The simulator therefore has no “true” Pacejka coefficients to recover, and the reference tire curve of Section VI is itself a Magic Formula fit to the simulator’s own per-wheel force, slip and load telemetry under a dedicated limit-excitation manoeuvre.

Three rows of Table I are load-bearing for the identification. First, the mass the physics engine simulates (269.6 kg, measured as total static normal load) is 12 % above the configured

TABLE II
ROS 2 INTERFACE USED BY THE IDENTIFICATION AND VALIDATION PIPELINE. NAMESPACE IS `/sim` UNLESS MARKED ABSOLUTE.

Topic	Type	Role
<code>/odom</code>	<code>nav_msgs/Odometry</code>	State v_x, v_y, ω ; twist is body-frame
<code>feedback/steering_angle</code>	<code>std_msgs/Float32 (deg)</code>	Achieved road-wheel angle δ
<code>feedback/imu</code>	<code>sensor_msgs/Imu</code>	a_x, a_y for the friction warm-start
<code>feedback/tire_forces</code>	<code>sim_manager_msgs/TireForces</code>	Per-wheel α, F_z, F_y ; ground truth
<code>feedback/speed</code>	<code>std_msgs/Float32</code>	Forward speed echo
<code>feedback/motors</code>	<code>std_msgs/String (JSON)</code>	Per-wheel speed/torque/brake
<code>/drive</code>	<code>ackermann_msgs/AckermannDriveStamped</code>	Control command in
<code>control/tire_friction</code>	<code>std_msgs/Float32</code>	Runtime grip override
<code>/clock (abs.)</code>	<code>roscpp_msgs/Clock</code>	Simulation time base

240 kg; since (7) is linear in m , the configured value biases every recovered force by that amount. Second, the PhysX wheel material friction coefficient (1.5) is *not* the realised peak axle friction (1.00–1.05), so an identifier tuned to converge on the configuration file converges on the wrong target. Third, the achieved road-wheel angle is 0.76 of the commanded angle, so slip angles built from the command are systematically too large.

C. CARLA_ASU_BRIDGE: interfaces and telemetry

CARLA_ASU_BRIDGE [28] is a C++ ROS 2 lifecycle node linked directly against CARLA’s client library (Fig. 2). It spawns and configures the ego vehicle, drives its inputs, and republishes simulator state as standard ROS 2 topics; all identification traffic crosses this interface.

1) *Command path:* The bridge runs in `ackermann_drive` mode, subscribing to a single `ackermann_msgs/AckermannDriveStamped` on `/drive` and forwarding speed and steering angle to CARLA’s native Ackermann controller, whose server-side cascaded PID loops are configured once at startup. No client-side steering PID is present—the commanded road-wheel angle passes through in radians—so the achieved/commanded ratio of Table I is a property of the vehicle and its steering curve rather than of a tuning choice in the bridge.

2) *Feedback path:* Table II lists the interface consumed by the identifier and its validation. Three properties are specific to this architecture.

a) *Per-wheel ground truth:* `feedback/tire_forces` is a direct passthrough of CARLA’s own `WheelTelemetryData`: per-wheel lateral slip angle, normal load, lateral force, longitudinal force and wheel torque, in wheel order [FL, FR, RL, RR]. No analytic tire model exists inside the bridge, so nothing on this topic can disagree with the physics the vehicle is driven by. Lateral force is sign-flipped and slip angle converted into the ROS right-handed body frame; the longitudinal channel is drivetrain torque at the axle ($-\tau/r_w$) rather than a contact-patch force

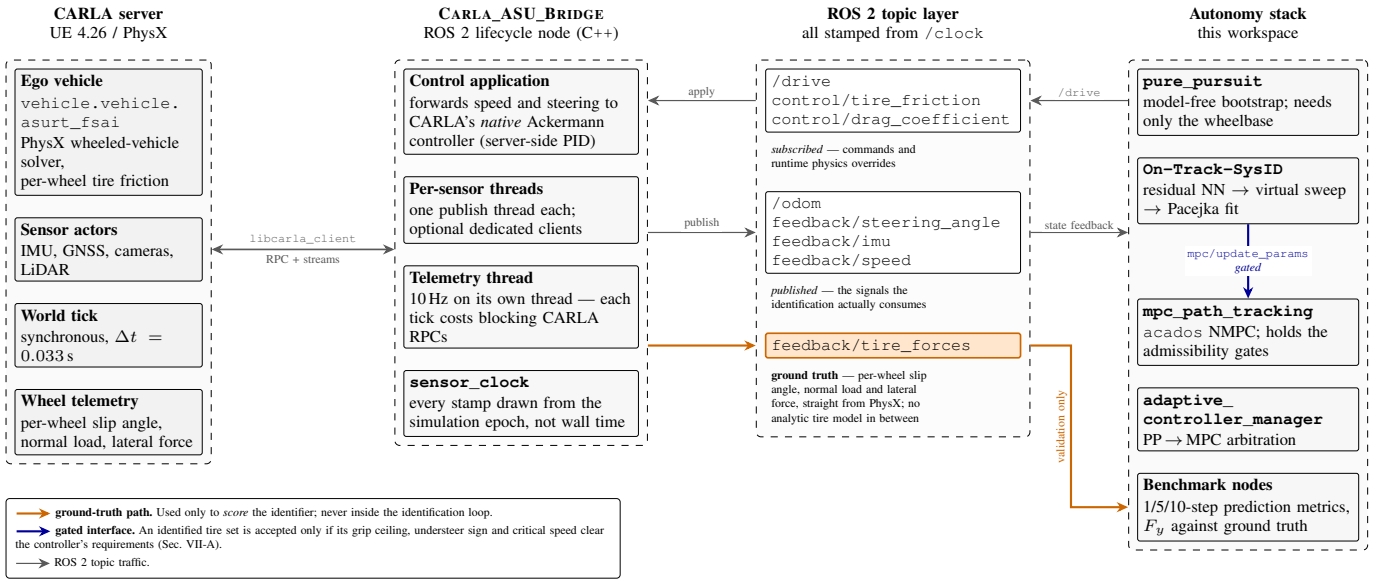


Fig. 2. Architecture of the CARLA_ASU_BRIDGE ROS 2 interface. The bridge links against CARLA's C++ client, drives the ego vehicle through CARLA's native Ackermann controller, and republishes state, sensors and per-wheel telemetry on ROS 2 topics stamped from the simulation clock. The `feedback/tire_forces` path (highlighted) carries the simulator's own per-wheel slip angle, normal load and lateral force, and is the ground truth against which the identifier is scored.

and is unused here. This topic is what allows an on-track identifier to be scored against the forces the plant generated.

b) Simulation-time stamping: Every feedback message is stamped from the same simulation epoch as `/clock`. In synchronous mode CARLA ticks as fast as the server manages (observed near $1.9\times$ real time), so wall-clock stamping would drift the identification dataset against itself without bound.

c) Odometry and frame conventions: Odometry is computed from the actor transform and velocity at 50 Hz with `follow_server_rate` enabled and an optional additive Gaussian noise model (disabled here). Publisher and subscribers all use `BEST_EFFORT`, because a reliability mismatch on this interface is silent: DDS requires requested $\leq$ offered, so a `RELIABLE` request against a `BEST_EFFORT` publisher yields one warning and then no data. Because (3) is meaningless in the wrong frame, the body-frame assumption on `/odom`'s twist was verified rather than assumed: correlation 0.93 with 0.09 m/s RMS error against the reconstructed body-frame lateral velocity, versus 16.7 m/s RMS under a world-frame interpretation.

3) Extensions made for identification: Four changes support this study and are listed for reproducibility: the existing but unsubscribed `feedback/steering_angle` is now consumed by the identifier (with configurable units, scale and staleness timeout, falling back to the `/drive` setpoint); runtime overrides `control/tire_friction` and `control/drag_coefficient` write straight to the live `VehiclePhysicsControl`, so surface friction can change within a run without respawning; the torque curve, zero-throttle damping and wheel `tire_friction` were tuned against a low-speed hold test (a wheel friction above ≈ 2.0 on a steered wheel scrubs hard enough at creep speed to stall the drivetrain intermittently), and these values define the plant being identified; and telemetry runs on a dedicated

10 Hz thread off the control loop, since each tick costs several blocking CARLA RPCs.

D. Data acquisition and preprocessing

The vehicle is driven around the reference raceline by a model-free pure-pursuit controller, which requires no tire knowledge—only the wheelbase—and is therefore a valid bootstrap, as in the baseline [1], [4]. Odometry, the achieved steering angle and the drive command are sampled into a ring buffer at 50 Hz ($T_s = 0.02$ s) for 30 s, gated on $v_x > 1$ m/s. Preprocessing follows the baseline: a zero-phase low-pass filter, then symmetry augmentation by mirroring $(v_y, \omega, \delta) \rightarrow (-v_y, -\omega, -\delta)$ at unchanged v_x , which doubles the data and removes the cornering bias of a circuit. Re-identification retriggers every 30 s by default.

E. How the architecture closes the loop

The four stages of the identification pipeline map onto the architecture as follows. *Acquisition* uses `/odom`, `feedback/steering_angle` and `/drive`, all stamped on the simulation clock, so that the sampling interval used by the one-step model is the interval the physics actually advanced. *Preprocessing* and *fitting* run inside the identification node against that buffer. *Validation* draws on `feedback/tire_forces` and `feedback/imu`, which never enter the identification loop, so the estimator is scored against a channel it did not observe. *Deployment* pushes the fitted coefficients over a ROS 2 service to the controller behind the gates of Section VII. Only the last stage leaves the architecture, which is what makes the negative result of Section VI-G a statement about the trajectory rather than about the interface.

Loop diagram of the four identification stages, with this paper's changes highlighted.

Fig. 3. The identification pipeline. Recorded lap data trains the residual network; the corrected model is rolled out through a synthetic steering sweep; Pacejka coefficients are fitted to that sweep; the fit becomes the next nominal model. Highlighted: the excitation-aware rollout speed (Section V-B), the extrapolation-bounded sweep and frozen regularisation target (Section V-C), and the admissibility gate on the output (Section VII).

V. IDENTIFICATION METHODOLOGY

This section describes the identification pipeline as it stands after correction. Section V-A covers the residual model, V-B–V-C the two changes to the virtual-data step that constitute the paper's central methodological contribution, V-D the fit itself, V-E the measurement chain, and V-F the validation protocol. Figure 3 summarises the loop.

A. Residual model

The default residual is the baseline multilayer perceptron: 4 inputs, one hidden layer of 8 units, 2 outputs, LeakyReLU, 58 parameters, full-batch Adam at 5×10^{-4} on an MSE objective. Because the full-scale residual has different structure from the scaled platform's, the architecture is selectable: `baseline`, `wide`, `physics_inputs` (slip angles appended to the input vector), `ensemble`, and `s4`—an S4D diagonal state-space temporal residual [15], [16] evaluated by causal recurrent scan over a sliding window of 20 samples (0.4 s, matched to tire-relaxation timescales). As stated in Section II, the S4-family residual overlaps with concurrent work [14] and is not claimed as a contribution. An optional physics-informed loss [29], [13]—dynamics consistency, left/right symmetry and input-gradient smoothness—is available as an ablation switch; the results below use the baseline MSE objective.

B. Excitation-aware virtual rollout

This is the central correction. Consider stage 2 of Section III-D: a constant-speed rollout at v with the steering ramped to δ_{sweep} . In quasi-steady cornering the lateral acceleration a single-track vehicle develops kinematically is $a_y = v^2 \kappa$ with $\kappa \approx \delta/L$, so the largest friction coefficient the rollout can demand of the tire is

$$\mu_{\text{reach}} \approx \frac{v^2 \delta_{\text{sweep}}}{g L} \quad (8)$$

Equation (8) contains no tire parameter: it is a property of the rollout configuration alone. If $\mu_{\text{reach}} < D$, the synthetic data never leaves the linear ramp of (4), only the product BCD is

identifiable (Section III-C), and the bounded optimiser resolves the residual freedom on a box edge—precisely the signature reported in Section VI-B.

The baseline implementation selects the rollout speed as $v = \text{clip}(\bar{v}_x, 2, 4)$ m/s, a value representative of a 1:10 platform. On our vehicle, with $\delta_{\text{sweep}} = 0.4$ rad and $L = 1.5334$ m, (8) gives $\mu_{\text{reach}} = 0.43$ at 4 m/s: the rollout cannot ask the tire for more than 0.43 g whatever the tire is. At 1:10 scale, with $L \approx 0.32$ m, the same rollout speed yields $\mu_{\text{reach}} \approx 2.0$ and the pathology is invisible.

1) *Correction*: The rollout speed is now taken from the speed the vehicle was actually driven at, clamped by configuration bounds (lower bound 7 m/s, no upper bound), and the pipeline warns whenever (8) evaluates below the prior's D —turning a silent degeneracy into a startup diagnostic. The same bound is the right tool for choosing a data-collection speed: identification data should be gathered at $v \geq \sqrt{gLD/\delta_{\text{sweep}}}$.

C. Bounded extrapolation and frozen-prior regularisation

Raising the rollout speed exposes a second, opposite failure. The rollout queries the residual network at steering angles it has never seen: on a pure-pursuit lap the observed $|\delta|$ stays below ≈ 0.06 rad while the sweep ran to 0.4 rad. The fit then reads the network's extrapolation as additional grip, and because each iteration's answer became both the next iteration's nominal model *and* its regularisation target, the error compounds: D walked from the 1.009/1.002 prior to 1.5402/1.8721 over six iterations, a claimed 1.7 g on a vehicle measuring 1.0 g in steady state.

Two changes close the loop:

- 1) **Extrapolation bound.** The sweep terminates at $\delta_{\text{sweep}} = \text{clip}(\lambda_e \cdot P_{99}(|\delta_{\text{obs}}|), \delta_{\text{min}}, \delta_{\text{max}})$ with $\lambda_e = 1.5$, $\delta_{\text{min}} = 0.05$ rad, $\delta_{\text{max}} = 0.34$ rad, so the network is never queried more than 50 % beyond the steering range it was trained on.
- 2) **Frozen prior.** The Tikhonov target is frozen at the configured prior Φ_p^0 for all iterations, while the optimiser's *start point* continues to track the previous iteration. The per-axle objective is

$$\min_{\phi} \sum_k \rho(r_k(\phi)) + \sum_j w_j (\phi_j - \phi_j^0)^2, \quad (9)$$

with r_k the force residual, ρ a soft- ℓ_1 robust loss and w_j scaled to the data residual budget by a single dimensionless weight $\lambda = 0.05$. The regulariser exists specifically to hold the coefficients the manoeuvre does not excite— E above all—at their nominal value instead of on a box edge.

D. Pacejka fit

Fitting applies the steady-state force recovery of (7) to the synthetic data under box constraints $\Phi_p \in ([4.0, 1.2, 0.4, -3.0], [20.0, 2.2, 2.0, 1.0])$ per axle. Relative to the baseline we (i) raised the number of jittered multi-starts from 1 to 8, (ii) added a derivative-free simplex (Nelder–Mead) and a global differential-evolution solver as alternatives to trust-region reflective, and (iii) corrected the configured prior,

which previously initialised D below its own lower bound and thus started every optimisation on the rail it then failed to leave. The prior now carries the measured tire of Table III.

None of (i)–(iii) fixes the degeneracy on its own, and they were verified as secondary: with the rollout speed left at 4 m/s, a synthetic $D = 1.5$ tire fits back as $D_f = 0.753$, $D_r = 0.737$ with three coefficients railed regardless of solver or start point; at 8–20 m/s the same tire fits back as $D = 1.40$ – 1.61 with nothing railed.

E. Measurement-side corrections

Three corrections apply to the measurement chain rather than to the estimator.

1) *Achieved versus commanded steering*: α_f in (3) was built from the /drive setpoint. The measured achieved/commanded ratio is 0.76, so α_f was systematically $\approx 25\%$ too large and the fitted front stiffness $\approx 20\%$ too small (13725 N/rad against the simulator’s 17198 N/rad on the same run). Using feedback/steering_angle recovers 18486 N/rad.

2) *Mass and geometry*: m and l_{wb} are taken from the measured static wheel loads (Table I) rather than from the configuration file, and $l_{wb} \equiv l_f + l_r$ is enforced by a regression test, because (7) uses the wheelbase for the normal loads and the axle split for the force division; the two disagreeing is a silent bias.

3) *Friction warm-start as a floor*: An initial estimate of D is derived from measured accelerations, in the spirit of model-free friction estimation [17]. The baseline-adjacent implementation computed the median of $\sqrt{a_x^2 + a_y^2}/g$ over an explicitly low-slip mask, which is structurally wrong: it measures utilised friction over samples selected for being far from the limit, and utilisation lower-bounds capability. On a 13 m/s pure-pursuit lap the median measures 0.044 against a peak of 0.46. The estimator is now a 99th percentile over the whole buffer, applied as a floor on the prior’s D , never as a replacement, and only on the first identification of a node lifetime. This is a corrected implementation detail rather than a criticism of the friction-estimation literature, whose methods target available friction and assume near-limit excitation.

F. Validation protocol

Three levels of validation are used, in increasing strength:

- 1) **One-step prediction** on a held-out run, as in the baseline: RMSE and R^2 of v_y and ω under $\hat{\mathbf{x}}_{k+1} = f(\mathbf{x}_k, \mathbf{u}_k, \Phi_p)$. A dedicated ROS 2 node extends this to configurable multi-step horizons (1/5/10 steps) with Euler/Heun/RK4 integration and online Welford statistics.
- 2) **Direct force comparison** against feedback/tire_forces: the identified curve is evaluated at the measured (α, F_z) and compared per axle against the measured F_y . This level is unavailable on a physical platform without a force-measuring hub and is the strongest evidence in the paper.
- 3) **Closed-loop tracking** under the MPC using the identified model: lap time, mean and maximum lateral

TABLE III
REFERENCE TIRE MODEL OF THE PLANT, OBTAINED BY FITTING THE MAGIC FORMULA DIRECTLY TO THE SIMULATOR’S PER-WHEEL TELEMETRY OVER A DEDICATED LIMIT-EXCITATION RUN REACHING $\alpha = 0.37$ RAD, AXLE $\mu = 1.05$ AND 1.50 g. DIFFERENTIAL EVOLUTION, E CONSTRAINED TO ≥ -2 .

Axle	B	C	D	E	RMSE
Front	7.076	1.346	1.009	−2.000	135 N
Rear	7.873	1.383	1.002	−1.024	85 N

deviation, steering sign reversals per control step, and RMS steering rate against the rate limit. This level is not completed here; see Section VI-G.

The steady-state force recovery of (7) was itself validated against the simulator’s axle force on a normal lap (correlation 0.993, slope 0.94). This matters because (7) is not a force measurement—it is fully determined by $v_x\omega$ —so its adequacy had to be established rather than assumed.

VI. EXPERIMENTS AND RESULTS

A. Setup

All experiments run in simulation on the architecture of Section IV: the Formula AI vehicle in CARLA [8], driven through CARLA_ASU_BRIDGE on a ROS 2 Humble workspace, with the CARLA server in synchronous mode at a 0.033 s fixed physics step. Identification data is collected over 30 s of pure-pursuit driving at 50 Hz, matching the baseline’s data budget [1]. Ground truth for forces, slip angles and normal loads is the simulator’s own per-wheel telemetry.

1) *Reference trajectory*: Every run follows the same optimised raceline (Fig. 4), produced with the TUM global_racetrack_optimization package [30] under its minimum-curvature formulation [24] on a recorded centreline of the CARLA silverstone circuit: a closed 5830 m lap of 2916 waypoints at 2.0 m spacing, reference speed 14.90–46.02 m/s, minimum corner radius 18.7 m ($|\kappa|_{\max} = 0.0536/\text{m}$).

Two of the optimiser’s inputs matter for Section VI-G. Its vehicle configuration assumes a mass of 204 kg against the 269.6 kg the physics engine simulates, and its ggV envelope declares a constant lateral capability of 12.0 m/s² at every speed. The resulting profile saturates that envelope exactly—peak demand $\max(v_x^2|\kappa|)$ over the lap is 12.00 m/s², i.e. 1.22 g—so the trajectory is feasible by construction for the vehicle the optimiser was told about, and only for that vehicle. Against the plant’s measured steady-state ceiling of ≈ 9.8 m/s², 11.9 % of the lap is infeasible (Fig. 4c).

Table III is the reference against which the on-track identifier is scored. Note $D \approx 1.0$, not the 1.5 of the wheel material coefficient—the distinction of Section IV-B.

B. The failure mode, reproduced

Running the baseline pipeline unmodified on this platform yields

$$\begin{aligned} \text{C_Pf} &= [5.2874, 1.5536, \mathbf{0.4}, \mathbf{1.0}], \\ \text{C_Pr} &= [19.3236, \mathbf{1.2}, 0.4046, -0.1292], \end{aligned}$$

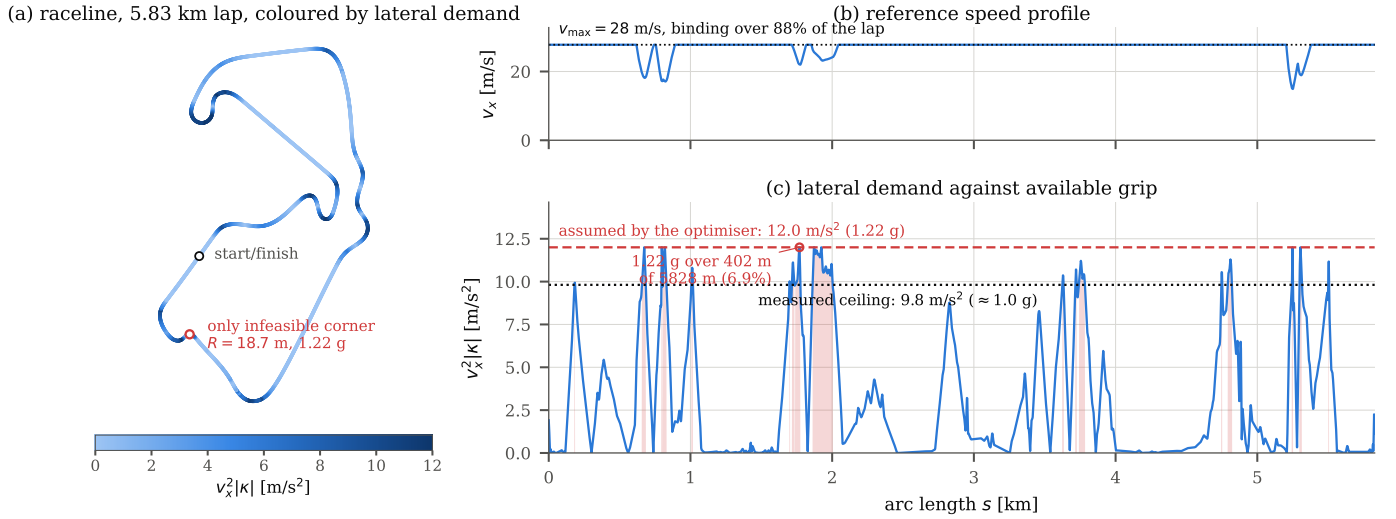


Fig. 4. The optimised raceline used for every experiment, generated with the TUM `global_racetrajectory_optimization` package [30] under its minimum-curvature formulation [24] from a recorded centreline of the CARLA silverstone circuit. (a) Track layout, coloured by reference speed; the marked corner ($R = 18.7$ m) sets the feasibility limit. (b) Reference speed profile, 14.90–46.02 m/s. (c) Lateral demand $v_x^2|\kappa|$ along the lap. The profile saturates the constant 12.0 m/s² envelope given to the optimiser (dashed) but not the grip this vehicle has (dotted, ≈ 1.0 g, measured); the shaded segments, 11.9% of the lap, demand more than the plant can deliver. This is the trajectory the corrected identifier’s admissibility gate rejects.

TABLE IV

CLOSED-LOOP CONSEQUENCE OF THE DEGENERATE TIRE SET, MEASURED IN A HARNESS DRIVING THE RACELINE FOR 20 s WITH THE PURE-PURSUIT→MPC SWITCH AT $t = 5$ s.

Controller model	Max cross-track	Infeasible stages
Plausible prior ($D = 1.5$)	0.03 m	0
Identified set @ 11.1 m/s	0.16 m	0
Identified set @ 15 m/s	0.43 m	0
Identified set @ 27 m/s	41.89 m	29 461

with the bold entries exactly on their box constraints and B_r just under its upper bound: five of eight coefficients on or against a rail. The implied physics is qualitatively wrong, not merely inaccurate—a grip ceiling of 3.95 m/s² (0.40 g) and a front axle stiffness of 4012 N/rad, so the model believes it needs ≈ 0.30 rad of front slip to generate 1 g, beyond the vehicle’s 0.2793 rad steering limit. The model believes the car cannot corner. Table IV shows why this stays invisible until it is not: lateral demand scales as v^2 .

C. Isolating the cause

Four hypotheses were tested against measurement rather than argued from code. Table V records the outcome, including the two that were wrong; the eliminated hypotheses are as informative as the confirmed one.

The decisive experiment is an offline round-trip with the residual network zeroed, so that only the rollout is under test: a synthetic $D = 1.5$ tire, rolled out at 4 m/s and re-fitted, returns $D_f = 0.753$, $D_r = 0.737$ with three coefficients railed; the same tire rolled out at 8–20 m/s returns $D = 1.40$ –1.61 with nothing railed. Since the tire is known exactly and the residual model is absent, this isolates the rollout configuration as the cause and confirms (8). Figure 14 shows the round trip as a sweep over rollout speed, next to the reachable-friction bound

TABLE V

HYPOTHESES TESTED AGAINST THE SIMULATOR’S OWN TELEMETRY.

Hypothesis	Verdict
Friction warm-start measures utilised friction over a low-slip mask	<i>Structurally confirmed, not the cause.</i> Utilisation measures 0.044 median, 0.46 peak at 13 m/s. Corrected anyway (Sec. V-E).
Static D prior below its own lower bound	<i>Confirmed, secondary.</i> Corrected; multi-starts 1→8.
Only BCD identifiable at low slip	<i>Confirmed, and it is the mechanism</i> —but caused by the rollout speed, not by the recorded lap (Sec. V-B).
/odom twist in world frame	<i>Refuted.</i> corr = 0.93, RMS 0.09 m/s against a body-frame interpretation.
Steady-state force model (7) inadequate	<i>Refuted as a cause.</i> corr = 0.993, slope 0.94 against the simulator’s own axle force.
δ taken from command, not achieved	<i>Confirmed, significant.</i> Ratio 0.76; worth 20% of front stiffness.

that predicts it and the slip-angle coverage each configuration delivers to the fit.

D. Baseline versus this work

Table VI places the pipeline as published [1] beside the pipeline as it stands here, on the axes that changed. The residual model, the data budget, the four-stage loop and the steady-state force recovery are inherited unchanged; what differs is the configuration of the virtual-data step, the regularisation topology, the measurement chain, and the contract at the output.

E. Identification after correction

With the six defects of Table VII corrected, three consecutive live identifications return no coefficient on a bound. Peak friction coefficients land at $D_f, D_r \in [1.05, 1.18]$ against the reference 1.009/1.002 of Table III; axle cornering stiffnesses land at $C_f \approx 21$ kN/rad and $C_r \approx 20$ kN/rad against the simulator’s measured 17–19 kN/rad; the understeer gradient

TABLE VI

THE BASELINE PIPELINE [1] AGAINST THIS WORK, ON THE AXES THAT DIFFER. ROWS MARKED *inherited* ARE UNCHANGED AND ARE LISTED SO THAT THE EXTENT OF THE MODIFICATION IS UNAMBIGUOUS.

Axis	Baseline [1]	This work
Platform	1:10, 3.7 kg, $L = 0.32$ m	Full scale, 269.6 kg, $L = 1.53$ m
Rollout speed	$\text{clip}(\bar{v}_x, 2, 4)$ m/s	From the measured lap, floored at 7 m/s
Reachable μ	≈ 2.0 (adequate)	0.43 under the baseline clip; ≥ 1.2 after correction
Sweep amplitude	Fixed 0.4 rad	$\text{clip}(1.5P_{99}(\delta_{\text{obs}}), \delta_{\min}, \delta_{\max})$
Regularisation	None; each fit is the next start point	Tikhonov to a <i>frozen</i> prior; start point still tracks
Steering signal	Command	Achieved road-wheel angle
Mass, geometry	Configuration file	Measured static wheel loads
Friction prior	—	P_{99} of measured $ a /g$, as a floor
Solver	Trust-region, single start	Multi-start, plus simplex and global options
Output contract	Per-coefficient box constraints	Gates on derived quantities at the controller interface
Residual model	MLP, 58 parameters	<i>inherited</i> (with selectable alternatives)
Data budget	30 s, one controller-free lap	<i>inherited</i>
Force recovery	Quasi-steady yaw balance	<i>inherited</i>
Ground truth	Steady-state manoeuvre in a large space	Simulator per-wheel force, slip and load

has the correct sign ($l_f C_f < l_r C_r$) on every run, matching the measured plant; and one-step prediction on held-out data gives $R^2 = 0.97$ for v_y and $R^2 = 0.99$ for ω . The peak-friction drift across the six co-identification iterations is removed: where the unbounded sweep walked D from 1.01 to 1.87, the bounded sweep with a frozen prior settles inside $[1.05, 1.18]$.

F. Ablations

Two ablation axes are exposed by the implementation and are marked as not yet run rather than reported with placeholder numbers. The first is the residual architecture (baseline MLP, wide, physics_inputs, ensemble, s4), scored on one-step and 10-step RMSE and on the identified coefficients; the relevant comparison is [14], which reports a lateral-force RMSE reduction exceeding 60 % for an S4 residual, and we expect a smaller effect here because our sequences are short and our residual is dominated by a correctable measurement bias rather than by unmodelled temporal dynamics. The second is the loss mode (mse versus physics_informed) on the same dataset, comparing one-step metrics, per-iteration coefficient trajectories and closed-loop tracking.

G. Negative result: the raceline, not the model

With the identifier corrected, level 3 of the validation protocol (Section V-F) does *not* complete, and the reason is instructive. The identified model reports a grip ceiling of 1.02–1.16 g; the plant sustains ≈ 1.0 g in steady state; the raceline demands $\max(v_x^2|\kappa|) = 12.00 \text{ m/s}^2 = 1.22 \text{ g}$ ($P_{99} = 11.90 \text{ m/s}^2$). The admissibility gate of Section VII therefore rejects every identification, and does so correctly: the trajectory is infeasible for this vehicle, and the corrected identifier is what made that visible.

Six stacked panels sharing a 30 s time axis.

Fig. 5. Example telemetry from one identification run. (a) v_x ; (b) v_y and ω ; (c) commanded versus achieved road-wheel angle, whose ratio measures 0.76; (d) a_x, a_y ; (e) per-axle lateral force from the simulator’s wheel telemetry against the value recovered by the steady-state relation (7); (f) per-axle slip angle.

μ_{reach} versus rollout speed, one curve per sweep amplitude, with the old clip band and the plant’s measured μ .

Fig. 6. Reachable friction of the synthetic rollout, $\mu_{\text{reach}} = v^2 \delta_{\text{sweep}} / (gL)$, for the full-scale vehicle ($L = 1.53$ m, solid) and a 1:10 platform ($L = 0.32$ m, dashed). The shaded band is the rollout speed inherited from the scaled platform; within it the full-scale rollout cannot demand more than $\mu = 0.43$, so the tire’s peak factor is not identifiable from it. The same configuration is comfortably above the peak at 1:10, which is why the defect is scale-latent.

The cause is worth naming because any model-based stack can inherit it. The trajectory optimiser was handed a `ggv` envelope asserting a constant 12.0 m/s^2 of lateral capability and a vehicle mass of 204 kg, and it did its job—the profile saturates that envelope to the last digit. The envelope was not this vehicle’s. Identification and trajectory generation were parameterised from two different descriptions of the same car, and nothing in the stack compared them until the gate did.

We do not loosen the gate. The quantitative consequence for regeneration is that the tightest corner, $|\kappa| = 0.0536/\text{m}$, permits $\sqrt{g\mu/|\kappa|} = 13.5 \text{ m/s}$ at the measured grip (13.7–14.6 m/s at the identified ceiling of 1.02–1.16 g); capping the speed profile at 13 m/s brings peak demand to 9.05 m/s^2 (0.92 g), whereas 15 m/s returns it to 12.00 m/s^2 . Regenerating the raceline for the *measured* vehicle ($m = 269.6 \text{ kg}$, $\mu_{\text{peak}} = 1.0$, $v_{\text{max}} \approx 13\text{--}14 \text{ m/s}$) is the prerequisite for the closed-loop experiment and is the immediate next step (Section IX).

VII. INTEGRATION WITH THE CONTROLLER

The identified model leaves the pipeline through one interface. A nonlinear MPC path-tracking controller (`acados` [22]),

TABLE VII

DEFECTS IDENTIFIED, CORRECTED AND REGRESSION-TESTED IN THIS WORK. THE EFFECT COLUMN REPORTS THE MEASURED CHANGE WHERE ONE WAS QUANTIFIED.

Defect	Root cause	Correction	Measured effect
Degenerate fit, D on its bound	Pacejka fit runs on a synthetic rollout whose speed was clipped to 2–4 m/s; $\mu_{\text{reach}} = 0.43$	Rollout speed from measured lap speed, floored at 7 m/s; warning when $(8) < D$	D : 0.40 (railed) $\rightarrow$ 1.05–1.18; C_f : 4.0 $\rightarrow$ 21 kN/rad
Identified set never reached the controller	Trainer wrote coefficients to file but returned nothing; the node re-read a static prior file and submitted <i>that</i>	Trainer returns the identified coefficients; node submits them	Service received the prior verbatim while the fit had produced a different set
D drifts upward across iterations	Sweep extrapolated the residual network far past its training range, and each answer became its own regularisation target	Sweep bounded at $1.5 \times P_{99}(\delta)$; regularisation target frozen at the configured prior	1.01 $\rightarrow$ 1.87 drift removed; settles 1.05–1.18
Front stiffness underestimated	Slip angle built from the steering <i>setpoint</i>	Achieved road-wheel angle from feedback/steering_angle, with fallback	C_f : 13725 $\rightarrow$ 18486 N/rad (truth 17198)
Normal loads 10–15% low	m copied from the configuration file (240 kg) against a simulated 269.6 kg; $l_{wb} \neq l_f + l_r$	Measured mass and geometry; $l_{wb} \equiv l_f + l_r$ enforced by test	$F_{z,f}/F_{z,r}$: 1222/1134 $\rightarrow$ 1345.6/1298.9 N
Friction prior pinned near 0.4	Median of <i>utilised</i> friction over an explicitly <i>low-slip</i> mask	P_{99} over the whole buffer, applied as a floor on the prior	Median 0.044 vs peak 0.46 at 13 m/s

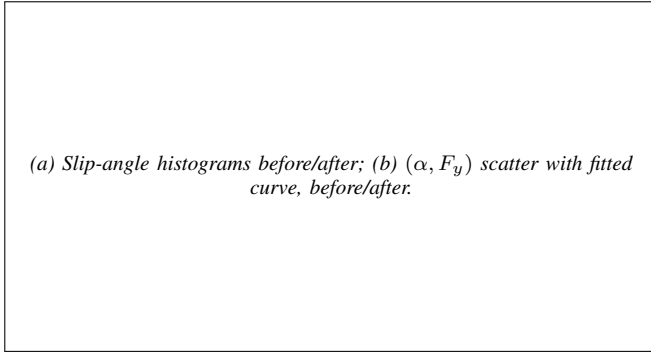


Fig. 7. Excitation entering the Pacejka fit. (a) Distribution of the slip angles present in the synthetic data before and after the rollout correction, with the reference tire’s peak-force slip angle marked. (b) The corresponding (α, F_y) scatter with the fitted curve; before the correction all data lies on the initial linear ramp, where only the product BCD is identifiable.

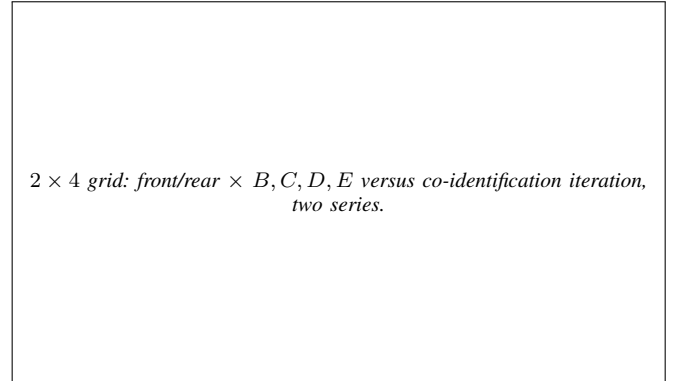


Fig. 8. Coefficient trajectories over the six co-identification iterations, with the unbounded sweep and tracking regularisation target (which walks D from 1.01 to 1.87) against the bounded sweep with a frozen target (which settles in [1.05, 1.18]). Dashed lines mark the box constraints and the reference values of Table III.

partial-condensing HPIPM [23], horizon $N = 100$ at a 50 Hz control period) carries the same single-track model as the identifier and consumes Φ_p directly: the eight Pacejka coefficients enter the prediction model’s axle forces through (4), and nothing else about the controller changes when a new identification arrives. Coefficients are pushed over a ROS 2 service at the end of each identification cycle, so the controller’s model tracks the plant at the re-identification rate. We claim no contribution to the controller design; it appears here as the consumer that defines what an admissible identification is.

A. Admissibility gates

A coefficient set can lie strictly inside every box constraint and still be unusable. We observed three distinct ways, each requiring a gate on a *derived* quantity rather than on a

coefficient:

- 1) **Grip ceiling versus trajectory demand.** The controller computes $\max_k (v_{x,k}^2 / |\kappa_k|)$ over the speed-clamped reference and rejects any identification whose $a_y^{\max}$ from (6) falls below it. The degenerate fit reported a 0.40 g ceiling against a 1.22 g demand and produced 41.89 m of cross-track error at 27 m/s, against 0.03 m for a physically plausible set (Table IV).
- 2) **Understeer sign and critical speed.** A set with $l_f C_f > l_r C_r$ whose v_{crit} falls inside the controller’s speed range makes the prediction model open-loop unstable, and the resulting $\rho(A_d) > 1$ is raised to ρ^N in the condensed QP. An observed set gave $C_f = 37031$ N/rad against $C_r = 11691$ N/rad, i.e. $v_{\text{crit}} = 17.4$ m/s.
- 3) **Integration stiffness.** Even a stable set can be stiff

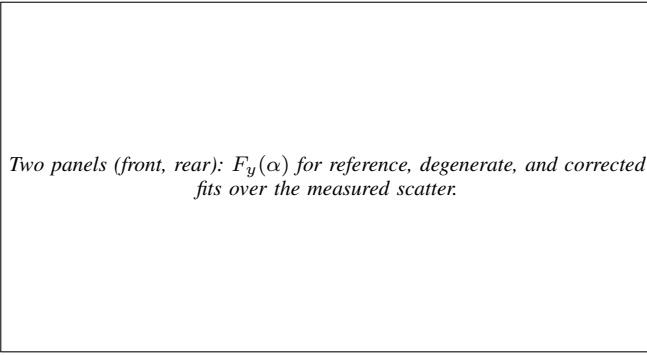


Fig. 9. Identified tire models against the plant. Each panel shows the reference fit of Table III, the degenerate result of the untransplanted pipeline, and the corrected result over three consecutive identifications (shaded band), with the raw per-wheel (α , F_y) measurements underneath.

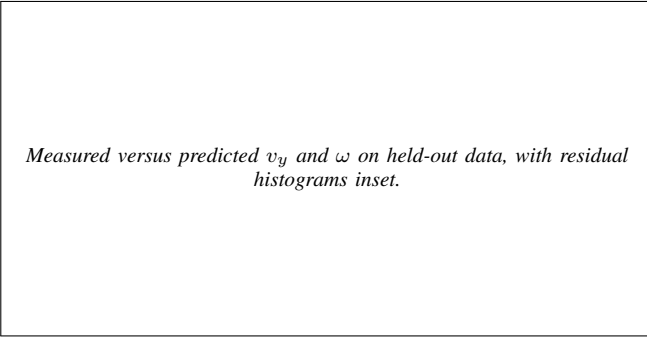


Fig. 10. One-step prediction on a held-out run using the corrected identified model: $R^2 = 0.97$ for lateral velocity and $R^2 = 0.99$ for yaw rate.

relative to the control period: a single explicit RK4 step at $dt = 0.05$ s gave $\rho(A_d) = 547$ at 5 m/s for one identified set. The model sub-steps adaptively, sizing the sub-step from the model's own fastest lateral mode $|\lambda| \approx (C_f + C_r)/(mv_x) + (l_f^2 C_f + l_r^2 C_r)/(I_z v_x)$ so that $|\lambda|h \leq 1.5$, restoring $\rho \approx 1.000$ across the speed range at 2.3–6.8 ms per 100-stage horizon.

The gates are the contract between identification and control: an identification that passes is a model the controller can be held responsible for, and one that fails is reported as unidentified rather than silently deployed. Two further controller-side changes were required before any identified model could be evaluated at full-scale speeds and are recorded for completeness rather than as identification results: reference speeds are clamped at ingest, and the solved state is rolled forward by the measured transport delay before each solve, which took cross-track error at 27 m/s from 130.55 m to 0.38 m.

VIII. DISCUSSION AND LIMITATIONS

A. What generalises

The reachable-friction bound (8) is kinematic: it involves no tire parameter, no simulator-specific quantity and no property of the residual model, so it applies to any implementation of the virtual-steady-state-data strategy of [1], in simulation or on hardware. Its practical content is a design rule—the *rollout must be configured to demand at least the friction one intends to identify*—and, where it cannot, D should be reported as

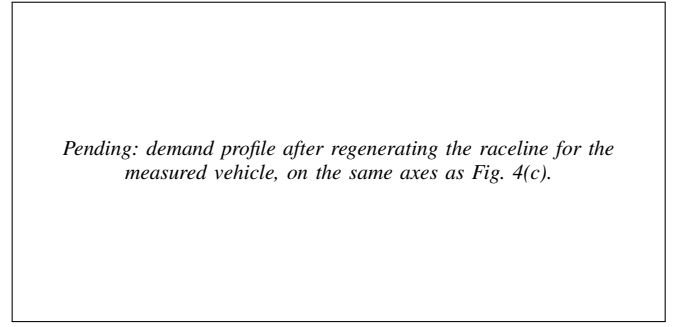


Fig. 11. Regenerated speed profile. **Not yet produced:** the demand-versus-grip comparison for the *current* trajectory is Fig. 4(c); this figure is its counterpart after the raceline is regenerated for the measured vehicle ($m = 269.6$ kg, $\mu_{\text{peak}} = 1.0$), which is the prerequisite for the closed-loop experiment of Fig. 13.

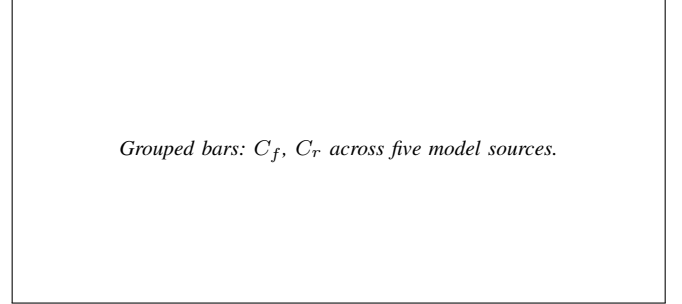


Fig. 12. Axle cornering stiffness $C = F_z B C^{\text{MF}} D$ across model sources: simulator ground truth, the degenerate fit, a fit referenced to the steering command, a fit referenced to the achieved steering angle, and the final corrected fit. The understeer condition $l_f C_f$ versus $l_r C_r$ is annotated per group.

unidentified rather than as fitted. The diagnostic signature, coefficients resting exactly on box constraints, is likewise generic and should be checked and logged by any bounded tire-fitting routine, since it is otherwise indistinguishable in the output from a successful fit.

The insufficiency of per-coefficient box constraints for deployment admissibility is also platform-independent. Both failures we encountered—a set that satisfies every box constraint while being degenerate, and a set that satisfies every box constraint while being pairwise oversteering and unstable within the controller's operating range—are properties of the parameterisation, not of CARLA. Deep Dynamics [13] constrains coefficients inside the network; we argue that a second, derived-quantity check belongs at the interface where the model is handed to a controller.

B. What does not generalise

Three results are properties of this plant and must not be read as statements about tires. First, the numerical tire values: the plant is a PhysX wheeled-vehicle model with a configured friction coefficient, so Table III describes *that* plant, and agreement with it is evidence that the identifier recovers the plant it observes, not evidence about real tire behaviour. Sagmeister *et al.* [19] show that sensitivity to model fidelity is largest near the acceleration limits, which is where this study operates, and R-CARLA [18] exists because

Pending: cross-track error versus time, and driven path over the raceline.

Fig. 13. Closed-loop tracking under MPC with the identified model. **Not yet produced:** this experiment is blocked on regenerating a feasible raceline for the measured vehicle, see Section VI-G.

CARLA’s native dynamics are a known weak point for racing-grade work; hardware replication is therefore necessary before any claim about identification accuracy on a real vehicle. Second, the gap between configured wheel friction (1.5) and realised peak axle friction (1.00–1.05) is a property of how the PhysX solver composes friction with load and slip; we report it only because an identifier tuned to converge on the number in the configuration file will converge on the wrong number. Third, the 0.76 achieved/commanded steering ratio arises from CARLA’s Ackermann controller and this vehicle’s steering configuration—the lesson, identify against the achieved actuator state, is general and standard practice; the magnitude is not.

C. Limitations

- 1) **Closed-loop validation is incomplete.** Level 3 of Section V-F has not been run end to end, because the admissibility gate correctly rejects the model for the raceline in hand (Section VI-G). Until a feasible raceline is regenerated this paper reports open-loop accuracy and closed-loop *consequence* (Table IV), not closed-loop racing performance. We consider stating this preferable to relaxing the gate.
- 2) **No noise sweep.** The baseline’s robustness claim—a $3.3\times$ RMSE advantage over nonlinear least squares under injected noise [1]—has not been reproduced here. The bridge’s odometry noise model exists and is disabled for these runs, so the experiment is available; it has not been run, and our results are therefore not directly comparable to the baseline’s robustness figures.
- 3) **No same-data NLS baseline.** We compare against the plant’s own telemetry rather than against a nonlinear least-squares identification of the same data. That comparison would strengthen the paper and is straightforward to add.
- 4) **Single surface, vehicle and trajectory.** All results come from one vehicle configuration on one map. The runtime friction override supports a surface-change adaptation experiment—arguably the strongest argument for on-track identification—but it is not performed here.
- 5) **Yaw inertia is assumed, not measured** (Section A).
- 6) **Ablations not completed**, so no claim is made about the S4D residual or the physics-informed loss on this

platform (Section VI-F); in particular we do not claim to reproduce the S4 improvement of [14].

- 7) **Longitudinal dynamics are out of scope.** As in the baseline only lateral dynamics are identified; combined slip and load transfer are neglected, and the simulator’s longitudinal wheel force channel is drivetrain torque rather than a contact-patch force, so it is unusable as ground truth for a longitudinal extension without further work.

D. Threats to validity

The most serious internal threat is that several corrections of Table VII were applied together rather than factorially; only the rollout-speed correction was isolated, by the zeroed-residual round-trip of Section VI-C. The effects attributed to individual measurement-side corrections come from re-fitting the same recorded run with one change at a time, which controls for the data but not for interaction between changes; a full ablation would settle this and is inexpensive offline. The most serious external threat is the simulator’s tire model, discussed above; the mitigation available within this study is that every claim is scored against the plant’s own per-wheel forces rather than against a secondary model, so the identification result is at least not self-referential.

IX. CONCLUSION AND FUTURE WORK

This paper studied the identifiability of a learning-based on-track Pacejka identification pipeline [1] at full vehicle scale, using a CARLA simulation architecture that exposes the plant’s own per-wheel forces as ground truth. Transplanted unchanged from a 1:10 platform, the pipeline fails in a specific, diagnosable and correctable way. The failure is not a solver, bounds or tuning problem: it is a loss of excitation created by the pipeline’s own virtual-data step, whose reachable friction $\mu_{\text{reach}} = v^2 \delta_{\text{sweep}} / (gL)$ depends on the rollout configuration and not on the tire. A rollout speed appropriate at 1:10 caps μ_{reach} at 0.43 at full scale, below the plant’s peak, so the peak factor is not identifiable from the synthetic data and the bounded optimiser resolves the ambiguity on a box edge.

Correcting this—together with an extrapolation bound on the steering sweep, a frozen regularisation target, identification against the achieved rather than the commanded road-wheel angle, measured mass and geometry, and a peak-rather-than-median friction floor—produces identifications with no railed coefficient, peak friction coefficients of 1.05–1.18 against a measured 1.00–1.05, axle cornering stiffnesses within 10–20 % of the simulator’s own, correct understeer sign, and one-step R^2 of 0.97/0.99. We further argue that per-coefficient box constraints are not a sufficient admissibility criterion for handing a model to a model-based controller, and place gates on derived physical quantities at that interface instead. The corrected identifier’s first useful act was to reject the reference trajectory: the raceline in hand demands 1.22 g from a vehicle that sustains 1.0 g . We report this rather than relaxing the gate.

A. Future work

- a) *Immediate:*

Pending: three panels: (a) μ_{reach} versus rollout speed at full scale and 1:10; (b) round-trip identified D versus rollout speed under the baseline and the excitation-aware configuration; (c) the $(\alpha_f, F_{y,f})$ coverage each configuration hands to the fit.

Fig. 14. The baseline pipeline [1] against this work, with the residual network held at zero so that the rollout configuration is the only variable. **Not yet produced;** generation recipe in `figures/make_baseline_comparison.py`. (a) The friction the synthetic rollout can demand, (8): the baseline’s 2–4 m/s clip reaches $\mu = 0.43$ on the full-scale vehicle but ≈ 2.0 on the 1:10 platform it was designed for, which is why the defect is scale-latent. (b) A known tire rolled out and fitted back across rollout speeds: inside the baseline clip the peak factor lands on its box bound, above 7 m/s it is recovered. (c) The $(\alpha_f, F_{y,f})$ data the Pacejka fit actually sees under each configuration, against the true curve; the baseline configuration supplies only the initial linear ramp, where solely the product BCD is identifiable.

- 1) Regenerate the raceline for the *measured* vehicle ($m = 269.6 \text{ kg}$, $\mu_{\text{peak}} = 1.0$, $v_{\text{max}} \approx 13\text{--}14 \text{ m/s}$) and run the closed-loop validation: lap time, mean and maximum lateral deviation, and steering sign reversals per control step, against a hand-tuned prior model as control.
- 2) Run the noise sweep of [1] on this platform using the bridge’s odometry noise model, with a nonlinear least-squares identification of the same data as the comparison arm.
- 3) Complete the residual-architecture and loss-mode ablations of Section VI-F, and a factorial ablation over the six corrections of Table VII.
- 4) Measure I_z rather than assuming it (Section A).

b) *Beyond:*

- 1) **Surface-change adaptation.** The bridge changes tire friction at runtime without respawning, making a step-change-in-grip experiment cheap: change μ mid-run and measure the identifier’s convergence time and the controller’s transient. This is the scenario on-track identification exists for, and it is difficult to stage on hardware.
- 2) **Excitation-aware data collection.** Equation (8) inverts into a requirement on the manoeuvre. A controller that deliberately inserts bounded slalom or ramp-steer excitation when the identifier reports insufficient slip-angle coverage would move the pipeline from passive on-track identification toward active, safety-bounded on-track experiment design.
- 3) **Uncertainty-aware output.** The gates of Section VII-A are binary. Reporting a posterior over the coefficients—the Bayesian variational path already implemented alongside the deterministic solvers—would let the controller weight the model by its confidence rather than accept or reject it.
- 4) **Hardware replication.** The decisive test of every claim in Section VIII is a physical Formula Student vehicle. The identification code is unchanged between simulation and hardware; what changes is the availability of ground truth, which is what makes the simulation study a

necessary precursor rather than a substitute.

APPENDIX A ASSUMPTIONS AND REPRODUCIBILITY NOTES

The following are assumptions or design choices rather than measured facts. They are listed explicitly so that a reader can judge how far each result depends on them, and so that each can be discharged by a specific measurement.

A1. Yaw inertia I_z

$I_z = 51.1 \text{ kg m}^2$ is carried in the model file and is *not* measured. It enters (2) directly and therefore scales the yaw-rate residual the network is trained on, and it enters the stiffness-adaptive sub-stepping of Section VII-A through $l_f^2 C_f + l_r^2 C_r$ over I_z . *How to discharge:* apply a known yaw moment (a step steer at constant speed with the resulting axle forces read from `feedback/tire_forces`) and fit I_z from the measured $\dot{\omega}$; or read the value the physics engine is using directly from the actor’s inertia tensor.

A2. Centre-of-gravity split

$l_f = 0.738142 \text{ m}$, $l_r = 0.795362 \text{ m}$ are taken from the controller configuration and confirmed by the vehicle owner. The measured static load split (51.2/48.8 front/rear) implies a marginally different split than the geometry does (51.9/48.1) — a 1.4% disagreement. We have not resolved it, and it is below the level at which any conclusion in this paper would change.

A3. CG height and load transfer

$h_{\text{cg}} = 0.3 \text{ m}$ is carried in the model file and unused: the single-track model of Section III-A neglects load transfer, following the baseline. Normal loads are static. On a full-scale vehicle at 1 g this is a stronger approximation than at 1:10, and it is a candidate explanation for part of the residual the network absorbs.

A4. Steering-angle sign and units

`feedback/steering_angle` is published in degrees and is treated as the front road-wheel angle, positive left, matching the ROS body frame. It was verified to agree with the front-left wheel joint state to within 0.5 %. Left and right road wheels are not distinguished; the single-track δ is taken as this single reported value.

A5. Steady-state force recovery

Equation (7) is not a force measurement — it is fully determined by $v_x\omega$ — so it misattributes transients. It is applied only to the *synthetic*, quasi-steady rollout, where the steady-state assumption is constructed rather than hoped for. Its adequacy was nonetheless checked on measured data against the simulator’s own axle force (correlation 0.993, slope 0.94); we treat that as validating the relation, not as validating a steady-state assumption on recorded laps.

A6. The residual network is not the plant

The corrected model $\hat{\mathbf{x}}_{k+1} + \mathbf{e}_k$ is queried by the rollout at inputs the network has only partially seen. The extrapolation bound of Section V-C limits but does not eliminate this: the sweep still reaches $1.5\times$ the observed 99th percentile of $|\delta|$. The choice of 1.5 is empirical, selected because it was sufficient to reach the tire’s peak at the corrected rollout speed while removing the observed drift; it is not derived.

A7. Simulator determinism and timing

CARLA runs in synchronous mode with a 0.033 s physics step, but sim time advances faster than wall time (observed $\approx 1.9\times$ on our hardware). All identification data is stamped from the simulation clock. We assume the physics step is uniform and that dead-reckoned odometry samples (disabled here via `follow_server_rate`) do not enter the identification buffer.

A8. Software configuration of record

Identification and control workspace: ROS 2 Humble, Python 3.10 (required — ROS 2 Humble ships CPython 3.10 ABI extensions only), PyTorch with CUDA on an RTX 2080-class GPU. The bridge is built and sourced as a separate overlay on top of that workspace, so that its message packages (`sim_manager_msgs`) resolve for the ground-truth topics; CARLA server 0.9.16 / Unreal Engine 4.26; `acados` 0.5.5 with partial-condensing HPIPM. Identification runs at 50 Hz with a 30 s buffer, six co-identification iterations, 200 training epochs, full-batch Adam at 5×10^{-4} .

A9. Regression tests

Eight tests pin the identifiability signature (no fitted coefficient on a bound; $l_{wb} \equiv l_f + l_r$; the rollout speed derivation) and run under `colcon test`, so that the degenerate-fit signature cannot return silently. This is a reproducibility mechanism rather than a scientific result, but it is the mechanism that would catch a regression of the paper’s central finding.

APPENDIX B

REFERENCE IMPLEMENTATION

The identification package, the ROS 2 bridge and the controller stack used in this paper are available as `On-Track-SysID` (extended from [1]), `CARLA_ASU_BRIDGE` [28], and the accompanying MPC and benchmarking packages. The benchmarking packages compute one-step and multi-step prediction metrics online (RMSE, MAE, NRMSE, MaxAE, bias, standard deviation and R^2 by Welford’s algorithm), compare estimated against ground-truth per-wheel lateral forces, and record controller-switching behaviour, and were built for this study.

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The ON-TRACK-SYSID pipeline studied here originates from the ForzaETH Race Stack and the work of Dikici *et al.* [1], released under an open-source licence; this paper reports a port, a correction, and an extension of that work rather than a reimplement from scratch.

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